

13.2 Gravitational Potential

Question Paper

Course	CIEA Level Physics
Section	13. Gravitational Fields
Topic	13.2 Gravitational Potential
Difficulty	Medium

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

Mars is considered to be an isolated sphere of radius R with its mass concentrated at its centre.

The variation of the gravitational potential ϕ with distance x from the centre of Mars is shown in Fig 1.1.

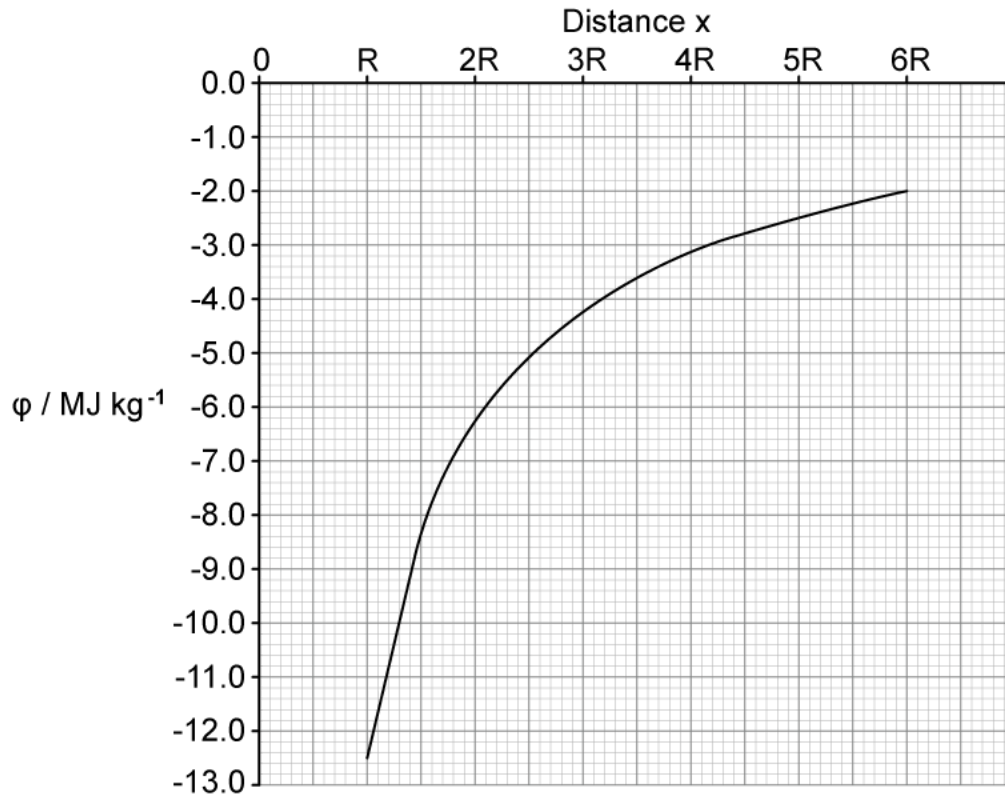


Fig 1.1

Explain why the values for ϕ in Fig 1.1. are negative.

[2 marks]

Question 1b

The radius R of Mars is 3400 km.

By considering the gravitational potential at Mars' surface, determine a value for its mass.

[3 marks]

Question 1c

A meteorite is at rest at infinity. The meteorite travels from infinity towards Mars.

Calculate the speed of the meteorite when it is at a distance of $3R$ above Mars' surface. Explain your working.

[4 marks]

Question 1d

Mars has two moons, Phobos and Deimos, as shown in Fig 1.2.

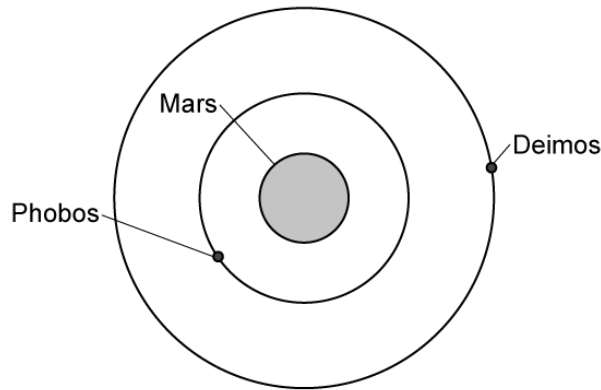


Fig 1.2

Calculate the work done to move a satellite of 210 kg from the orbit of Deimos to the orbit of Phobos.

Orbital radius of Phobos = 9400 km

Orbital radius of Deimos = 2.3×10^4 km

Work done = MJ

[3 marks]

Question 2a

The gravitational potential at a point A about the surface of a planet is -7.2 MJ kg^{-1} .

For a point B above the surface of the planet, the gravitational potential is -9.1 MJ kg^{-1} .

State, with a reason, whether point A or B is further away from the planet.

[2 marks]

Question 2b

A rocket is landing onto the surface of Mercury.

Table 1.1 gives data for the speed of the rocket at two heights about the surface of Mercury, after the rocket engine has been switched off.

Table 1.1

height / km	speed / m s^{-1}
$h_1 = 15 \times 10^3$	$v_1 = 7215$
$h_2 = 3.7 \times 10^3$	$v_2 = 7020$

Mercury may be assumed to be a uniform sphere of radius R , with its mass M concentrated at its centre. The rocket, after the engine has been switched off, has mass m . The change in gravitational potential energy of the rocket between the two heights is ΔE_p .

Show that the mass M of Mercury is given by the expression

$$M = \frac{\Delta E_p}{Gm \left(\frac{1}{R+h_1} - \frac{1}{R+h_2} \right)}$$

[3 marks]

Question 2c

Write an expression for the change in kinetic energy of the rocket.

[1 mark]

Question 3a

Venus may be assumed to be an isolated sphere of radius 6100 km with its mass, M , at 4.9×10^{24} kg concentrated at its centre.

An object is projected vertically from the surface of Venus so that it reaches an altitude of 7.2×10^4 km.

Explain why the equation $Mg\Delta h$ where g is the gravitational field strength of Venus and Δh is the height of the object, cannot be used to calculate the object's gravitational potential energy.

[1 mark]**Question 3b**

The change in the gravitational potential of a different object is 4.2 MJ kg^{-1} . This object ends up in orbit around Venus.

Calculate the radius of orbit of this object.

[3 marks]**Question 3c**

Calculate the change in work done per unit mass of the object projected vertically in part (a).

Change in work done per unit mass = MJ kg^{-1}

[2 marks]**Question 3d**

Show that the initial speed of the object in (a) is 1.1 km s^{-1} , assuming air resistance is negligible.

[3 marks]



Model Answers: Medium

1a

(a) The values for ϕ on Fig 1.1. are negative because:

- The gravitational potential (ϕ) is defined as 0 at infinity (∞); [1 mark]
- (The gravitational force is attractive so) the work is done by the field, so the potential at all points less than infinity is negative **OR** so the work must be done (on a mass) to reach infinity; [1 mark]

[Total: 2 marks]

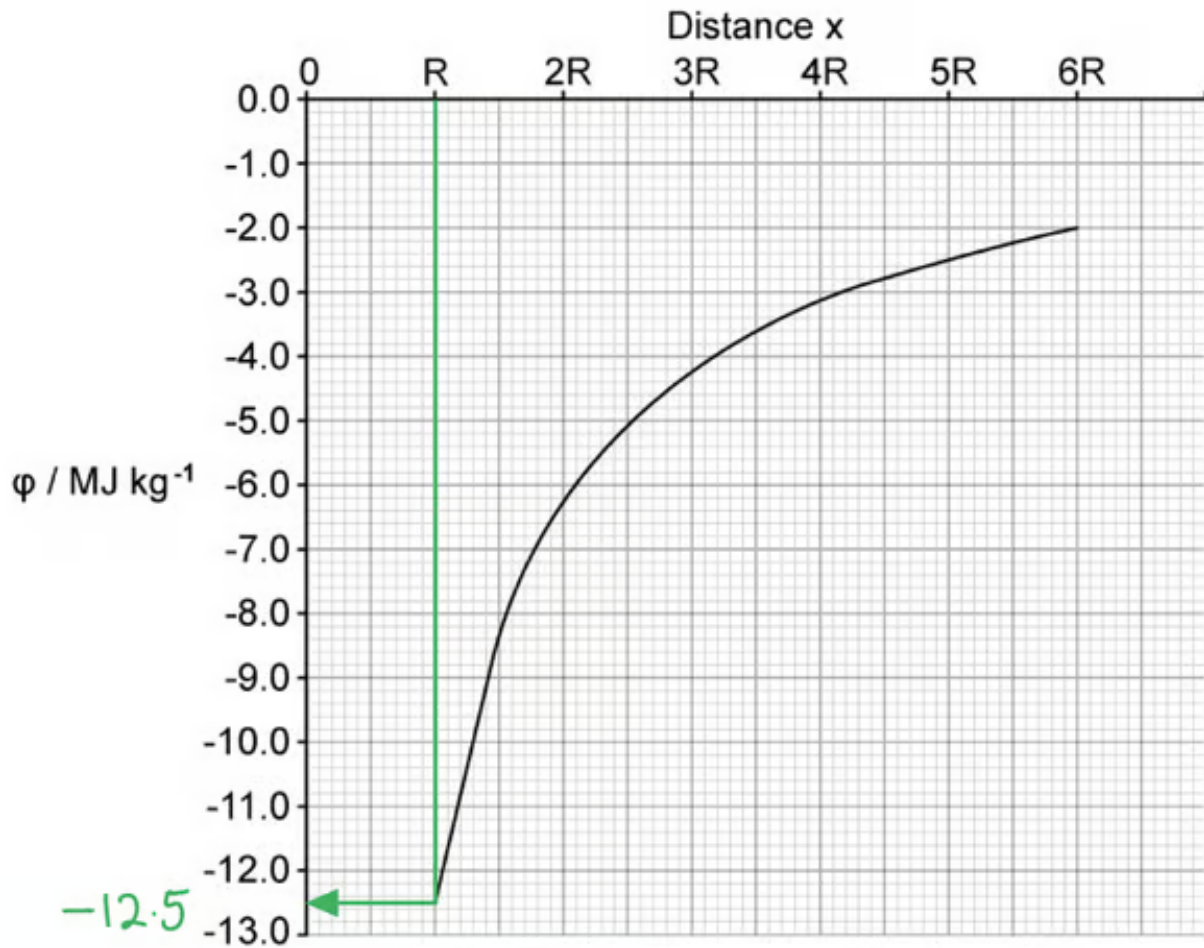
1b

(a) By considering the gravitational potential at Mars' surface, the mass of Mars is:

List the known quantities:

- Radius of Mars, $r = 3400 \text{ km} = 3400 \times 10^3 \text{ m}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Determine the potential at R , ϕ :



- $\phi = -12.5 \text{ MJ kg}^{-1} = -12.5 \times 10^6 \text{ J kg}^{-1}$ [1 mark]

Rearrange the gravitational potential equation for mass, M :

- $\phi = -\frac{GM}{r} \Rightarrow M = -\frac{r\phi}{G}$
- $M = -\frac{(3400 \times 10^3) \times (-12.5 \times 10^6)}{6.67 \times 10^{-11}}$ [1 mark]
- $M = 6.4 \times 10^{23} \text{ kg}$ [1 mark]

[Total: 3 marks]

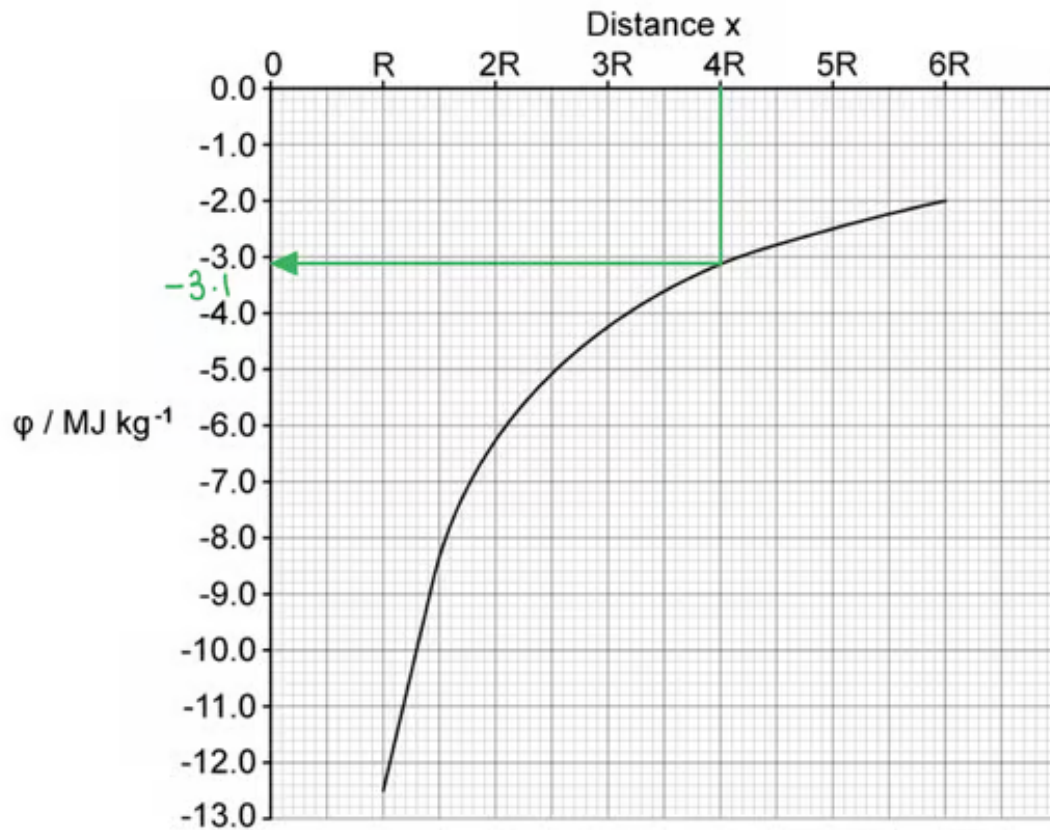
1c

(c) The speed of the meteorite when it is at a distance of $3R$ above Mars' surface is:

List the known quantities:

- Distance of meteorite above Mars' surface = $R + 3R = 4R$
- Mass of Mars, $M = 6.4 \times 10^{23} \text{ kg}$ (from part b)

Determine the loss in potential:



- $\phi = -3.1 \text{ MJ kg}^{-1} = -3.1 \times 10^6 \text{ J kg}^{-1}$ [1 mark]

State the energy conservation:

- Loss in gravitational potential energy = gain in kinetic energy [1 mark]

$$\bullet \quad -\frac{GMm}{r} = \phi m = \frac{1}{2}mv^2$$

Rearrange for the speed, v :

- $\phi = \frac{1}{2}v^2$
- $v = \sqrt{-2\phi}$

Substitute in the values:

- $v = \sqrt{-2 \times (-3.1 \times 10^6)}$ [1 mark]
- $v = 2.5 \times 10^3 \text{ m s}^{-1}$ [1 mark]

[Total: 4 marks]

Read the question carefully – it states we need to calculate the speed at the distance '3R above Mars' surface'. R is the distance to Mars' surface, so 3R above that is 4R. This is the distance at which we need to read the potential from Fig 1.1.

1d

(d) The work done to move a satellite of 210 kg from the orbit of Deimos to the orbit of Phobos is:

List the known quantities:

- Mass of the satellite, $m = 210 \text{ kg}$
- Mass of Mars, $M = 6.4 \times 10^{23} \text{ kg}$ (from part b)
- Orbital radius of Phobos, $r_p = 9400 \text{ km} = 9400 \times 10^3 \text{ m}$
- Orbital radius of Deimos, $r_d = 2.3 \times 10^4 \text{ km} = 2.3 \times 10^7 \text{ m}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

Calculate the change in gravitational potential energy:

- $E_p = -\frac{GMm}{r}$
- $\Delta E_p = -\frac{GMm}{r_p} - \left(-\frac{GMm}{r_d}\right) = \frac{GMm}{r_d} - \frac{GMm}{r_p}$ [1 mark]
- $\Delta E_p = GMm \left(\frac{1}{r_d} - \frac{1}{r_p} \right)$

Substitute in the values:

- $\Delta E_p = (6.67 \times 10^{-11}) \times (6.4 \times 10^{23}) \times 210 \left(\frac{1}{2.3 \times 10^7} - \frac{1}{9400 \times 10^3} \right)$ [1 mark]

- $\Delta E_p = -563.9 \times 10^6 = -560 \text{ MJ}$ [1 mark]

[Total: 3 marks]

The work done is negative, which is what is expected. From moving a satellite from the higher to lower orbit (as seen in Fig 1.2), work is done by the gravitational field, since there is an attractive force already towards Mars. This means the work done is **negative**, the satellite doesn't need to put work into the system to do this, it is already done for us!

If the satellite was to move from the orbit of Phobos to Deimos however, it would be moving away from Mars and away from the gravitational attractive force. Therefore, the work done would be **positive**.

2a

(a) The point further away from the planet is:

- point A (is further away); [1 mark]

Because:

Any **one** from:

- The potential increases as the distance from the planet increases (or vice versa); [1 mark]
- The potential is zero at infinity and at point A it is closer to zero; [1 mark]
- Potential $\propto \frac{1}{r}$ and at point A it is less negative / more positive; [1 mark]

[Total: 2 marks]

2b

(b) Show that the mass M of Mercury is given by the expression

$$\Delta E_p$$

- $$\Delta E_p = (0.01 \times 10^{-11}) \times (0.4 \times 10^{-22}) \times 210 \left(\frac{2.3 \times 10^7}{9400 \times 10^3} \right)^{11}$$

[1 mark]
- $$\Delta E_p = -563.9 \times 10^6 = -560 \text{ MJ}$$
 [1 mark]

[Total: 3 marks]

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2b

(b) Show that the mass M of Mercury is given by the expression

$$M = \frac{\Delta E_p}{Gm \left(\frac{1}{R+h_1} - \frac{1}{R+h_2} \right)} :$$

State the equation for the change in gravitational potential:

- $E_p = -\frac{GMm}{r}$
- $\Delta E_p = -\frac{GMm}{R+h_2} - \left(-\frac{GMm}{R+h_1} \right) = -\frac{GMm}{R+h_2} + \frac{GMm}{R+h_1}$ [1 mark]
- $\Delta E_p = \frac{GMm}{R+h_1} - \frac{GMm}{R+h_2} = GMm \left(\frac{1}{R+h_1} - \frac{1}{R+h_2} \right)$ [1 mark]

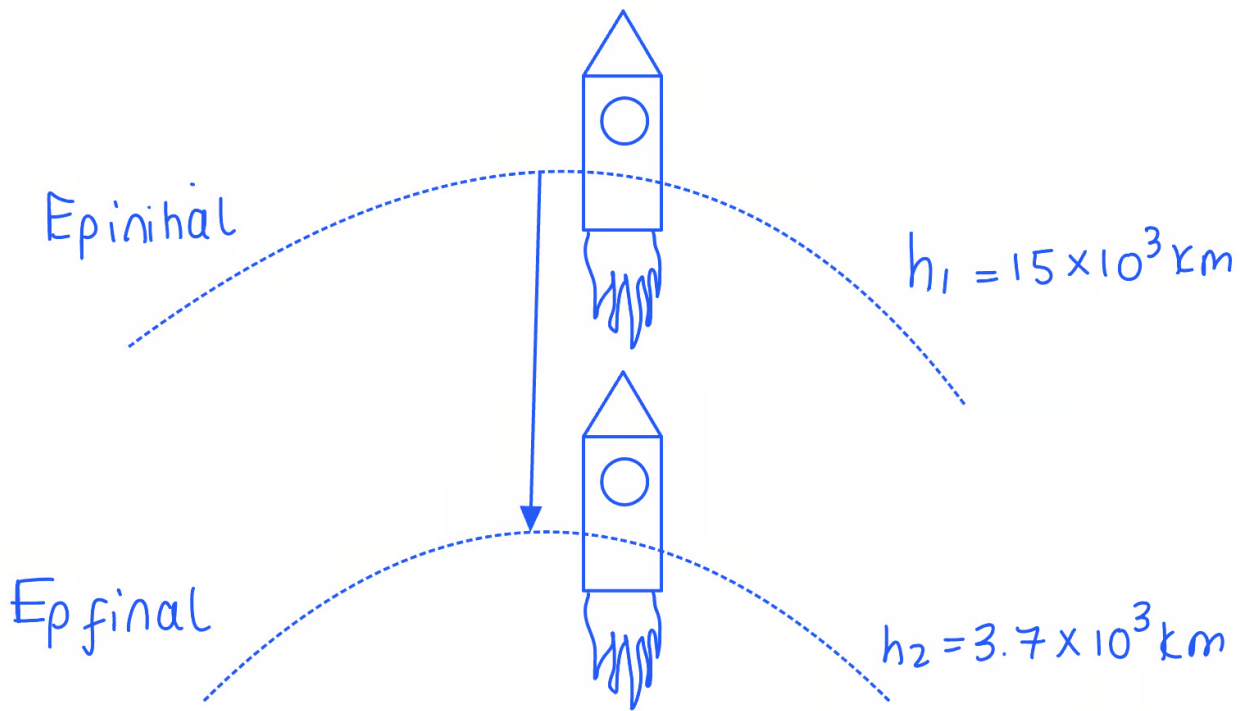
Rearrange for the mass of Mercury, M :

- $M = \frac{\Delta E_p}{Gm \left(\frac{1}{R+h_1} - \frac{1}{R+h_2} \right)}$ [1 mark]

[Total: 3 marks]

Remember that the r in the equation for gravitational potential energy is the distance from the **centre** of the planet creating the gravitational field. Therefore, the distance of the rocket from the centre is R + height.

The change in E_p is always the final E_p – initial E_p . Since the rocket is **landing** on Mercury, it is going from h_1 (the higher altitude) to h_2 (the lower altitude). This means the final E_p is at h_2 . Drawing a sketch can help to visualise this:



2c

(c) An expression for the change in kinetic energy is:

- $\frac{1}{2}m(v_2 - v_1)^2$ [1 mark]

[Total: 1 mark]

It doesn't matter here that $v_2 < v_1$, since the change in velocity is **squared**, $(v_2 - v_1)^2$ will still give a positive number.

The important thing to note is what is going on in this scenario. Since the rocket is landing, the change in kinetic energy is the **final** kinetic energy – **initial** kinetic energy.

3a

(a) The equation $Mg\Delta h$ cannot be used to calculate the gravitational potential energy because:

- The acceleration due to gravity (g) is not constant (at this altitude); [1 mark]

[Total: 1 mark]

3b

(b) The radius of orbit of this object is:

List the known quantities:

- Change in gravitational potential, $\Delta\phi = 4.2 \text{ MJ kg}^{-1} = 4.2 \times 10^6 \text{ J kg}^{-1}$
- Mass of Venus, $M = 4.9 \times 10^{24} \text{ kg}$
- Radius of Venus, $R = 6100 \text{ km} = 6100 \times 10^3 \text{ m}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

State the equation for the change in potential and rearrange for the radius of orbit, r .

$$\bullet \quad \Delta\phi = -\frac{GM}{r} - \left(-\frac{GM}{R}\right) = GM\left(\frac{1}{R} - \frac{1}{r}\right) \quad [1 \text{ mark}]$$

$$\bullet \quad r = \frac{1}{\frac{1}{R} - \frac{\Delta\phi}{GM}}$$

Substitute in the values:

$$\bullet \quad r = \frac{1}{\frac{1}{(6100 \times 10^3)} - \frac{4.2 \times 10^6}{(6.67 \times 10^{-11}) \times (4.9 \times 10^{24})}} \quad [1 \text{ mark}]$$

$$\bullet \quad r = 6.6 \times 10^6 \text{ m} \quad [1 \text{ mark}]$$

[Total: 3 marks]

3c

(c) The change in work done per unit mass of the object is:

List the known quantities:

- Mass of Venus, $M = 4.9 \times 10^{24} \text{ kg}$
- Radius of Venus, $R = 6100 \text{ km} = 6100 \times 10^3 \text{ m}$
- gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
- Altitude, $h = 7.2 \times 10^4 \text{ km}$

Determine the equation for the change in gravitational potential energy of the object, $\Delta\phi$:

$$\bullet \Delta\phi = -\frac{GM}{R+h} - \left(-\frac{GM}{R}\right) = GM\left(\frac{1}{R} - \frac{1}{R+h}\right)$$

Substitute in the values:

$$\bullet \Delta\phi = (6.67 \times 10^{-11}) \times (4.9 \times 10^{24}) \left(\frac{1}{6100 \times 10^3} - \frac{1}{(6100 \times 10^3) + (7.2 \times 10^4)} \right)$$

[1 mark]

$$\bullet \Delta\phi = 625.0 \times 10^3 \text{ J kg}^{-1} = 0.63 \text{ MJ kg}^{-1} \text{ [1 mark]}$$

[Total: 2 marks]

Remember the 'work done per unit mass' is another way of saying 'gravitational potential'. The exam question may not always ask this straight forwardly, so you should be familiar with the **definition** of all key terms in Physics.

3d

(d) Show that the initial speed of the object in **(a)** is 1.1 km s^{-1}

List the known quantities:

- Change in gravitational potential, $\Delta\phi = 625.0 \times 10^3 \text{ J kg}^{-1}$ (from part c)

Equate the gravitational potential energy and kinetic energy and rearrange for the speed, v :

$$\bullet \frac{1}{2}mv^2 = m\Delta\phi \text{ [1 mark]}$$

$$\bullet \frac{1}{2}v^2 = \Delta\phi$$

$$\bullet v = \sqrt{2 \times \Delta\phi}$$